TYCUS solutions

Service & Supply Catalog
2026

Dialysis Water & Machine Care

Reverse osmosis and hemodialysis machine maintenance, repair, calibration, parts, and supply and installation for dialysis units in the Philippines.



What we do

PM · Repair · Testing upon delivery · Calibration · Parts · Supply & installation

Equipment

Toray · SWS · Dialysis RO water systems

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1 About TYCUS Solutions

TYCUS Solutions is the trading name of TYCUS Enterprises Corporation, an instrumentation and control engineering company in Balayan, Batangas.

A dialysis unit stays safe on readings: pressures, flows, conductivity, temperature and alarm limits, plus the records that prove them. Measuring and controlling those values is what our engineers already do for industrial clients, and we bring the same discipline to your water system and your machines.



1.1 Credentials

Instrumentation & control engineering team

One team services the RO system and the machines it feeds, so nothing falls between two vendors.

Traceable calibration certificates

Meters and machine sensors are checked against traceable reference standards, with a certificate for your file.

PCAB-licensed contractor

Licensed for the installation work behind new RO systems, expansions and replacements.

Biomedical NC II holder technicians

Our technicians hold TESDA's Biomedical Equipment Servicing NC II.

1.2 How a new client starts

1

Free assessment

One visit and a written condition report of your water system and machines. You keep the report either way.

2

Baseline PM

Every stage and machine is read and recorded, so any drift is visible from the next visit on.

3

Agreed schedule

Monthly RO preventive maintenance, HD machines at manufacturer intervals, planned around your laboratory's visit.

Not sure where to begin? Tell us which matters more right now: keeping culture results low, or getting the maintenance file ready for inspection. We plan the first visit around that.

1.3 One partner for the water and the machines

In 30 seconds

TYCUS Solutions keeps your reverse osmosis system and your hemodialysis machines **maintained, documented and ready for inspection**. The same team carries out the monthly RO preventive maintenance, services and calibrates your Toray and SWS machines, supplies the parts, and installs new systems when you expand. You deal with **one accountable company for the water and the machines it feeds**, and your staff stay focused on patients.



1.4 What each client gets

Administrators & procurement

Concerns: cost overruns, audit findings, too many vendors.

One contract covers maintenance, calibration, reports and scheduled parts, so the water budget holds no surprises.

Nephrologists & head nurses

Concerns: pyrogenic reactions, infections, failed cultures, unreliable alarms.

Every treatment stage serviced and logged monthly, and every machine alarm tested at each visit.

Dialysis center owners

Concerns: lost sessions, LTO renewal, reputation.

A failed water test can close a unit for days. Regular PM and complete paperwork lower that risk.

1.5 Equipment we service

Equipment	Models
Toray hemodialysis machines	TR-8000 · TR-3300M · TR-3000MA · TR-3300S
SWS hemodialysis machines	SWS-4000 · SWS-4000A · SWS-6000
Dialysis RO water systems	Portable and central RO, with pretreatment, UV, ultrafilters and distribution loop

Other models and brands on request. Toray and SWS are trademarks of their respective owners; TYCUS Solutions is an independent service provider and is not affiliated with or authorized by either manufacturer.

2 Services

Six services, available on their own or combined in a service plan (page 11).

Service	What we do	You receive
Preventive maintenance	Monthly visits for the RO water system, as DOH requires, and HD machine PM at the manufacturer's intervals, recorded against your baseline.	PM report, readings and trend log, PM sticker with the next due date
RO & machine repair	Fault finding and repair of RO systems and HD machines, restoring them to specification, with the corrective action documented.	Diagnosis, quotation, repair report
Testing upon delivery	Every delivered unit or part is tested at your site before handover, and again after each PM or repair.	Signed pass/fail functionality test report
Calibration	Instruments and machine sensors calibrated against traceable reference standards.	Traceable calibration certificate issued by TYCUS*
Parts & consumables	Membranes, filters, media, UV lamps, sensors, seals and wear parts for Toray, SWS and RO systems, per item or bundled into a PM contract.	Quotation with lead time
Supply & installation	Complete RO systems sized to your stations, installed, commissioned and handed over ready for your DOH license application.	Drawings, manuals, test report, PM schedule

* TYCUS calibration is not ISO/IEC 17025 accredited. An accredited certificate is available on request.



Every visit ends on paper

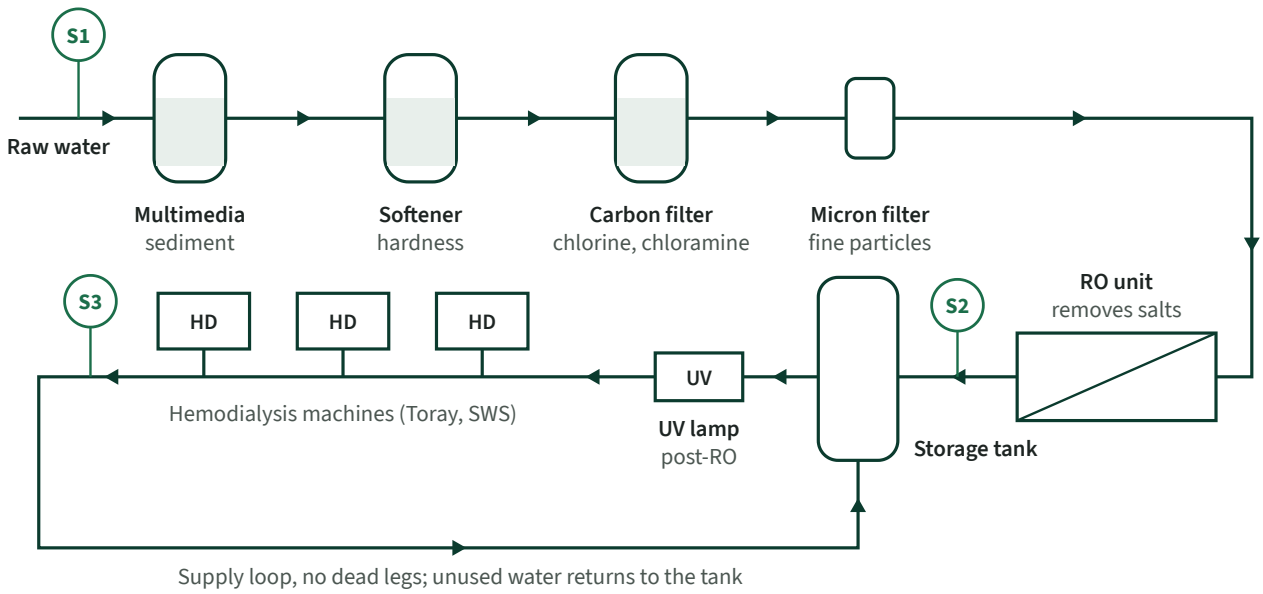
Inspectors ask for the maintenance file, and DOH expects the Medical Director to review and initial water test results. We leave the records ready for that review:

- PM report signed by the technician and your staff
- Functionality test report for each machine serviced
- Readings and trend log for the water system
- PM sticker with the next due date

3 Water system

We maintain the whole water path, from raw-water pretreatment to the dialysis machine, so it stays in the condition your laboratory's water tests depend on.

3.1 The water path we look after



S1–S3 DOH monthly sampling points: raw water, product water after RO, point of use.

All 8 treatment stages, plus the HD machines they feed, are serviced and logged at every monthly visit.

3.2 What we check, and what your laboratory tests

TYCUS on site, every monthly PM

- Total chlorine after carbon, against the 0.1 mg/L limit
- Hardness after the softener
- Product water conductivity / TDS, against a reference meter
- Pressures, flows, rejection and recovery
- Disinfectant residual after rinse

Your DOH-accredited laboratory

- Bacteria (heterotrophic plate count)
- Fecal coliform
- Endotoxin
- Chemical contaminants

TYCUS does not run a water analysis laboratory; we act on your laboratory's results.

3.3 Water quality limits, side by side

Parameter	DOH licensing limit AO 2013-0003	ISO 23500-3 limit / action level	Who tests, how often
Bacteria (HPC / TVC)	< 200 CFU/mL	**< 100 / 50 CFU/mL**	Accredited laboratory, monthly at raw, product and point of use
Fecal coliform	< 1.1 MPN/100 mL	**Covered by bacteria count**	Accredited laboratory, monthly
Endotoxin	Not specified in the AO	**< 0.25 / 0.125 EU/mL**	Recommended monthly with cultures
20 chemicals	AAMI maximums, e.g. free chlorine 0.5, chloramines 0.1, aluminum 0.01 mg/L	**Same contaminant list**	Accredited laboratory, before opening, then every 6 months at point of use
Product water TDS	Checked weekly; act above 10 mg/L	**Tracked as conductivity**	Clinic logbook weekly; TYCUS trends it at every PM

Sources: DOH AO 2013-0003; DOH AO 2006-0037; ISO 23500-3:2024; ISO limits as summarized in Clin J Am Soc Nephrol 2015;10(6):1061–1071.

3.4 When a water test fails

Under DOH AO 2013-0003, a unit whose result exceeds the limit stops operating and notifies patients until a repeat test complies. Our procedure works with your laboratory toward a passing retest, on record.

1. **Lab alert** — Result over the limit; the unit pauses per the DOH rule.
2. **TYCUS on site** — Priority call-out through our 24/7 support line.
3. **Fix the cause** — Clean or replace membranes; disinfect loop and machine hoses.
4. **Retest** — Re-sampling coordinated with your accredited laboratory.
5. **Reopen** — Laboratory confirms a pass; corrective action report for your DOH file.

Who takes the samples. TYCUS does not collect DOH water samples. In NCR and nearby provinces only DOH-accredited laboratory staff may sample, so we schedule PM around your laboratory's monthly visit.

3.5 Monthly RO preventive maintenance

Every stage is checked at every monthly visit and recorded against your baseline. Parts are replaced on your readings and the OEM manual, not on a fixed calendar.

Stage	Checked at every visit	Replaced when due
Multimedia filter	Pressure drop, backwash timer and valve	Media, as pressure drop and backwash show
Water softener	Hardness after softener, brine level and salt supply, regeneration settings	Resin, when outlet hardness rises
Carbon filter	Total chlorine after carbon, the worst-case point, checked against the 0.1 mg/L limit; backwash	Carbon media, before chlorine breaks through
Cartridge / micron	Pressure drop across housings	5 µm and 1 µm cartridges, on pressure drop
RO unit	Feed, permeate and reject pressures and flows; salt rejection and recovery against baseline; permeate conductivity against a reference meter; high-pressure pump noise, vibration, seals and motor current	Cleaned as trends demand; membranes replaced when performance falls
Storage tank & UV	UV lamp hours and intensity, quartz sleeve cleaned, tank vent filter	UV lamps and sleeves, by lamp hours
Ultrafilter & loop	Ultrafilter pressure drop; leaks, flow and dead legs; disinfection with a residual check after rinse	Ultrafilters, O-rings and gaskets; hoses as found
Controls & safety	Low-pressure switch, level controls, valves and alarms tested; wiring and grounding inspected	Switches and valves as found

3.6 Records for your maintenance file

After every visit you receive four records, ready for the Medical Director's initials and the DOH inspector's review.

1 **Signed service report**
Readings, parts replaced and next actions, on one page.

2 **Trend log**
Rejection, conductivity and pressures over time, so drift can show before a culture fails.

3 **Disinfection record**
Chemical used, contact time and a negative residual check after rinse.

4 **Parts traceability**
Part and lot numbers for membranes, cartridges and media.

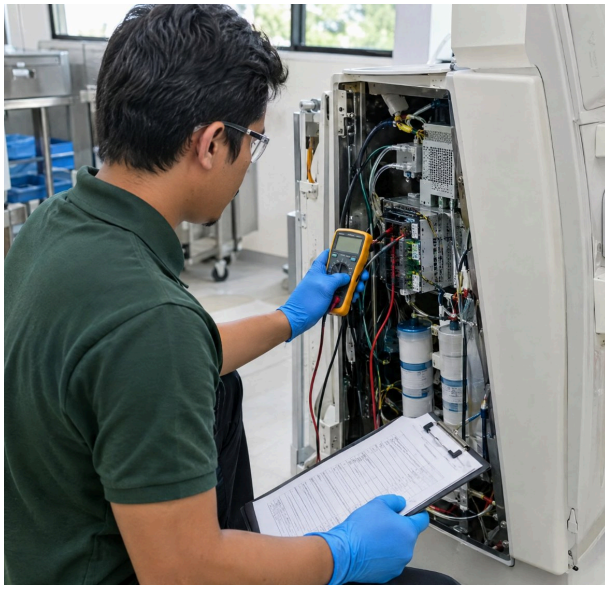
4 Hemodialysis machines

Preventive maintenance for Toray and SWS machines at every periodic visit, following the manufacturer's procedures, with readings checked against a calibrated meter.

A · Alarms every safety alarm triggered and confirmed

B · Sensors checked against a calibrated meter

C · Performance flows, UF and RO output measured



4.1 PM checklist

Self-test & records

- ☐ Machine self-test run; alarm and error logs reviewed
- ☐ Operating hours and service counters recorded

Hydraulics & dialysate

- ☐ Conductivity and temperature verified with a calibrated meter
- ☐ Ultrafiltration accuracy and balancing system checked
- ☐ Hydraulic leak check; dialysate filter replaced when due
- ☐ Heat and chemical disinfection cycles verified

Blood circuit

- ☐ Blood pump flow accuracy, rollers and occlusion
- ☐ Heparin pump delivery rate
- ☐ Arterial, venous and TMP pressure readings verified

Safety systems

- ☐ Air detector and blood leak detector tested
- ☐ Conductivity, temperature and pressure alarm limits
- ☐ Power-failure alarm and backup battery
- ☐ Electrical safety: protective earth and leakage current

Wear parts replaced per service interval

O-rings & seals | Dialysate filters | Pump segments & rollers | Filters & screens | Backup batteries

You receive a **PM report**, a **functionality test report** and a **PM sticker** with the next due date for each machine.

5 Supply & installation

For new units, expansions and end-of-life replacements: a turnkey RO system designed around your raw water and station count, tested at your site before the first patient treatment.

5.1 From survey to handover

1 Survey & raw water review
We check the feed water on site, review your laboratory's raw-water results and measure the room, then size the system to your stations plus planned growth.

2 Design to the DOH equipment list
Pretreatment, RO and storage, post-RO UV, and a loop laid out with no dead legs.

3 Inspect & install
Order checked on arrival; power, water and drain connected; sample ports fitted at raw, product and point of use.

4 Commission & test
Self-test and every safety alarm triggered; readings against a calibrated meter; flows measured; disinfection and flush, then sampling by your accredited laboratory.

5 Train & hand over
Test report signed with your team before first patient use; staff trained on daily checks and logbooks; drawings, manuals and PM schedule.

Required treatment train

DOH AO 2006-0037 equipment list

- ✓ Multimedia and bulk filters
- ✓ Carbon filter
- ✓ Water softener
- ✓ Reverse osmosis with storage tank

Every TYCUS design also includes **post-RO UV**.



Staff training at handover. Options to discuss: double-pass RO, heat-disinfectable loop, remote monitoring.

5.2 Your signed test report

Equipment ID & serial	Parameter tested	Reading	Acceptance limit	Pass / fail	Technician & date

The same test runs again after every PM or repair, and new installations move straight onto a monthly PM schedule from handover.

5.3 Parts & consumables

Genuine OEM or quality-equivalent parts, sourced on request. Order per item, or bundle scheduled parts into your PM contract.

Dialysis RO water systems

RO membrane elements	Toray 4040 / 8040; other brands too
Sediment & polishing cartridges	5 µm and 1 µm; 10" and 20"
Activated carbon	Granular media and carbon blocks
Softener resin & control valves	Resin refills, timer and meter valves
UV sterilizer parts	Lamps, quartz sleeves, ballasts
Ultrafilters	Endotoxin-retentive loop filters
Pumps & seal kits	High-pressure RO and booster pumps
Valves & switches	Solenoid valves, level switches
Housings & fittings	Membrane housings, FRP vessels, tubing

HD machines · Toray, SWS

Dialysate filters	Replaced by hours or treatments
Conductivity & temperature sensors	For dialysate monitoring
Pressure transducers	Arterial, venous and TMP
Blood & heparin pump parts	Rollers, segments, drive parts
Solenoid & check valves	Hydraulic circuit valves
O-ring & seal kits	Per service interval
Safety detector parts	Air and blood leak detectors
Power & backup	Power supply parts, backup batteries
Disinfectants & descalers	As specified by the manufacturer

5.4 Test & monitoring instruments

From our core instrumentation line, for your team's own daily checks between visits.

- Conductivity / TDS meters
- Inline conductivity monitors
- Chlorine & hardness test kits
- Pressure gauges & transmitters
- Flow meters
- Temperature sensors

6 Service plans

Three plans, each quoted to your equipment list and site. Most clients start with Scheduled Care.

Included	On-Call Repairs as needed	Most clients start here Scheduled Care Planned PM visits	Total Care One complete contract
Troubleshooting & repair	Yes	Yes	**Priority**
Testing upon delivery	Yes	Yes	Yes
RO system PM visits	–	Monthly	Monthly
HD machine PM visits	–	Manufacturer intervals	Manufacturer intervals
Signed reports & trend logs	Repair report	Yes	Yes
Traceable calibration	–	–	Yes
Parts & consumables	Per job	Per job	**Contract rates**
24/7 technical support line	Yes	Yes	Yes

Supply and installation is quoted per project. Repairs are quoted after evaluation; a fee of PHP 2,000.00 per unit applies if a unit assessed as repairable is not repaired.

6.1 Questions clinics ask us

Our technician already handles the RO.	Good. We support them with parts, trend analysis and a monthly report from a licensed service company, which is what DOH asks to see.
We already have a supplier.	Keep them for comparison. Ask both of us for last quarter's trend logs and see which would have warned you before a culture failed.
Is monthly PM worth the cost?	Compare it with the sessions lost if the unit stops for a retest: your stations, times sessions per day, times days closed.
Our water tests are passing.	Under 200 CFU/mL is the license line. Our trend logs flag falling performance early, so a bad month is less likely to turn into a shutdown.
Can we just call you when something breaks?	You can, but by then patients are being rescheduled. A monthly visit finds a fouling membrane or an exhausted carbon bed weeks earlier.

TYCUS solutions

Book a free RO system assessment

One visit, a written condition report of your water system and machines, and a PM proposal. You keep the report either way.

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References. DOH Administrative Order 2013-0003, Implementing Guidelines in the Analysis, Monitoring and Maintenance of Water Used in Dialysis Facilities. DOH Administrative Order 2006-0037, amendment to AO 2004-0163 on dialysis clinics. ISO 23500-3:2024, Water for haemodialysis and related therapies. Clin J Am Soc Nephrol 2015;10(6):1061–1071.

Toray and SWS are trademarks of their respective owners. Equipment under manufacturer warranty should be serviced per its warranty terms. TYCUS Solutions is a trade name of TYCUS Enterprises Corporation.